

sion course are taken through variable expansion, parameters, command substitution, quote removal, and so on. The result is also treated equal, the arithmetic expression is evaluated, and it may be nested. The evaluation of an arithmetic expression is performed as per the rules in shell arithmetic and if the expression is found invalid, then bash directly prints a message depicting standard error and no substitution of the value occurs.

Evaluation of this expression is done with the help of fixed-width integers having no check for the overflow. The division in this occurs by zero, which is usually recognised as an error. The operators are similar with C programming language and if the precedence decreases then the list appears as follows:

- VAR++ & VAR --
- ++VAR & -- VAR
- - and +
- ! And ~
- **
- *, / And %

Brace expansion – Brace expansion refers to mechanisms through which arbitrary strings can be generated. The patterns of this expansion are directly taken in the form of any optional segment of 'PERMEABLE' which is followed by a series of comma-separated strings present in between the pair of braces. These strings are followed by another optional 'POSTSCRIPT'. The PERMEABLE is easily prefixed to the strings contained within the pair of braces and postscript which is abundant to the resulting strings and expands them left to right. Brace expansion is nested where the result for every expanded string is not usually sorted, and the order is preserved from left to right.

Brace expansion mechanism is also performed before any other expansion, and all the characters which are special to other standards are preserved in it. This is also strictly textual, and is best not applied for any synthetic interpretation of the expansion context or for the text present between the braces. To eliminate any conflict with parameter expansion, \$ string is not considered equivalent to brace expansion. The correctly formed expansion essentially has unquoted opening or closing braces along with at least one unquoted comma. If the bash extension is formed incorrectly, then it will be left unchanged.

Parameter expansion – Parameter expansion is introduced by the character \$ and the symbol used in this is enclosed in braces (this is optional but it is used to protect the variable from expansion from the characters following it). Whenever the pressures are utilised then matching brace is the first '}', and any